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Cross-border venture capital performance: Evidence from China

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ABSTRACT

This paper investigates the determinants of cross-border venture capital (VC) performance in the Chinese VC market. We focus on the impact of foreign VC firms' (VCs') human capital and domestic entrepreneurs' experience on the performance of both VC investments and portfolio companies using logit and Cox hazard models. After controlling for portfolio company quality, domestic VC industry development, domestic exit conditions and a number of other factors, little correlation was evident between VC performance and foreign VCs' human capital, such as experience, networks and reputation. In contrast, the domestic entrepreneurs' experience is crucial to VC performance. In particular, if an entrepreneur has more general experience in terms of the number of companies previously worked for or more special experience in terms of the number of companies previously served as a CEO or top manager, a portfolio company is more likely to pull off a successful exit through IPO or M&A, and the VCs are also likely to shorten their investment duration in the portfolio company.

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1. Introduction

Since the founding of the American Research and Development Corporation in the U.S. in 1946, venture capital (VC) has experienced considerable growth. Many of today's multinational companies, such as Apple, Intel, Google and Microsoft, were funded by VC firms (VCs) in their early development stages. VCs are characterized as being "active investors" in financing startups in high-tech industries (Sahlman, 1990; Barry et al., 1990; Lerner, 1994; Kaplan and Strömberg, 2003). Besides providing funds, they add value by engaging in monitoring, recruiting executives and directors, formulating strategies, modernizing firms,

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Table 1

VC industries in China and the U.S. during 2002–2008. This table lists fundraising, the number of new funds, investment amount and the number of deals in the VC industries of China and the U.S. The data sources are Zero2IPO at <http://www.zero2ipo.com.cn> and the National VC Association at <http://www.nVca.org>.

Year	Fundraising (US\$M)		Number of new funds		Investment amount (US\$M)		Number of deals	
	China	U.S.	China	U.S.	China	U.S.	China	U.S.
2002	1298.0	3774.0	34	175	418.0	21,250.0	226	3105
2003	639.0	10,640.9	28	151	992.0	19,278.2	177	2952
2004	699.0	19,159.0	21	211	1269.0	22,117.5	253	3104
2005	4067.0	28,767.4	29	239	1173.0	22,922.2	228	3167
2006	3973.1	31,925.0	39	241	1777.42	26,334.4	324	3705
2007	5485.0	36,064.9	58	251	3247.05	30,639.3	440	3984
2008	7310.1	28,395.9	116	221	4210.47	28,076.9	607	3929

helping with fundraising activities and in other ways (Barry et al., 1990; Bottazzi et al., 2008). Such active involvement has been shown in theory to reduce information asymmetry and agency problems (Admati and Pfleiderer, 1994; Bergemann and Hege, 1998; Neher, 1999; Cornelli and Yosha, 2003; Casamatta, 2003; Schmidt, 2003; Wang and Zhou, 2004) and has been shown empirically to create public companies (Barry et al., 1990), enhance innovations (Kortum and Lerner, 2000), and improve performance (Bottazzi et al., 2008; Hochberg et al., 2007; Gompers et al., 2008).

VC funding in emerging markets is provided by either domestic or foreign VCs, but especially foreign ones. Cross-border VC investment means investments by foreign VCs in companies in other nations. Today's globalization of venture capital has been marked by an increasing number of cross-border entries of VCs from developed markets into emerging markets (Patricof, 1989; Sagari and Guidotti, 1992; Aylward, 1998). The entry of foreign VCs helps firms in emerging markets by shortening and reducing the costs of the learning process (Black and Gilson, 1998). Cross-border VC is essential in growth-oriented and high-risk new ventures in emerging markets where the domestic supply of VC is limited (Dossani and Kenney, 2002). Recognizing its importance, a number of developing countries have taken steps to stimulate and foster cross-border VC investments.

There have been an increasing number of studies of cross-border VC, but there is still much to learn about its investment strategies, monitoring mechanisms, exit methods, and short and long-run performance. To help fill this gap, this study is designed to investigate cross-border VC performance using Chinese data. The motivation for choosing China lies in the fact that China's VC market has experienced rapid expansion in recent years, and that it has the potential to expand much further. Also, foreign VCs dominate China's VC industry. China is now the largest net importer of VC funds, followed by Sweden, Canada, the UK, India, and France (Aizenman and Kendall, 2008).

China's own VC industry started in the mid 1980s. It experienced slow development in its first 10 years, but after the mid 1990s it grew rapidly and total investments have now reached No. 1 in Asia and No. 5 in the world (by some statistical rankings). An increasing number of well-known foreign VCs have now entered China, including Crystal Ventures, DCM-Doll Capital Management, Intel Capital, and JPMorgan Partners. Table 1, based on the China VC Report by Zero2IPO¹ and information from the U.S. National Venture Capital Association, compares the VC industries of China and the U.S. during 2002–2008, covering fundraising, the number of new funds, total investment amount and the number of deals. Table 1 indicates that China's VC industry has experienced explosive growth in the past several years, but its total size is still much smaller than that of its U.S. counterpart.

Even though foreign VCs constitute only about one quarter of China's VC industry in number, they dominate China's VC industry in many respects. Zeng (2004) has shown that most VC investments in China involve foreign VCs. According to the China VC Report by Zero2IPO, foreign VCs significantly outperform

¹ Zero2IPO is a well-known integrated service provider in the Chinese venture capital/private equity (VC/PE) industry. It has been conducting large-scale surveys and compiling a series of reports/directories on the VC industry since early 1999. The Zero2IPO database is China's first VC/PE investment industry database, which is also the only source for comprehensive and up-to-date information on VC/PE funds, firms, executives, and portfolio companies in China.

domestic VCs in fundraising, investment amount, the number of investment deals and portfolio company performance. In recent years, over 70% of total VC investment capital has been provided by foreign VCs, over 65% of VC funds that focus on China are managed by foreign VCs, and over 85% of new funds are raised by foreign VCs. In particular, a major proportion of successful deals that exit through IPOs or acquisitions are funded by foreign VCs. Many well-known, innovative, high-tech companies in China such as 51job, Shanda, Sina, Focus Media and China Financing Online received funding from foreign VCs in their seed or startup stages of development.

This study investigates the determinants of cross-border VC performance using a sample of 495 VC investments in 243 Chinese domestic companies. Most previous studies of VC performance have focused on domestic VCs, especially on U.S. VCs in the U.S. market. In contrast, foreign VC performance in China's domestic market is the focus of this study. All the VCs in our sample are foreign VCs. The performance of both VC investments and portfolio companies are investigated using logit and Cox hazard models. The impact of both foreign VCs' human capital and domestic entrepreneurs' experience on VC performance is studied.² Foreign VCs' human capital from VC markets worldwide, such as experience, networks and reputation, is found to have little effect on their performance in China. This finding is surprising, since VCs' human capital has been greatly emphasized in the literature as essential to their performance. We conjecture that cultural, institutional and capital market differences between established and emerging markets tend to render human capital somewhat market-specific. It has much less value in emerging markets.

On the other hand, we find that domestic entrepreneurs' experience is critically important to cross-border VC performance. Specifically, if a CEO has more general experience measured by the number of companies previously worked for or more special experience measured by the number of companies previously served as CEO or top manager,³ the domestic portfolio company is more likely to pull off a successful exit through an IPO or M&A, and VCs shorten their investment duration. This finding provides more evidence of the effect of a portfolio company's entrepreneurial experience on VC performance.

We also find that cross-border VC performance is hardly related to market conditions for IPO and M&A in the domestic market, but strongly related to the founder's departure, the separation of the CEO and founder, the general quality of a portfolio company, and the development of the domestic VC industry. We observe that in China many such investments exit through IPOs or acquisitions in overseas markets instead of the domestic market. This may explain why domestic market conditions for IPO and M&A have little effect on foreign VC performance.

The remainder of this paper is organized as follows. [Section 2](#) presents a literature review. [Section 3](#) describes data, variables and methodology. [Sections 4 and 5](#) analyze VC performance respectively at the VC investment level and the portfolio company level. [Section 6](#) conducts extensive robustness checks. Finally, [Section 7](#) concludes the paper.

2. Literature review

There are three related literatures to our study, which focus respectively on Asia–Pacific VC markets, cross-border VC investments, and VC performance. In the first related literature, there is an expanding literature on emerging VC markets, especially on Asia–Pacific countries. Asia–Pacific countries have grown rapidly and become the main VC importers since the late 1990s. Many studies have investigated the legal and institutional differences between emerging and established markets. [Cumming et al. \(2006\)](#) analyzed the impact of legality on VC exits among 468 VC-backed companies from 12 Asia–Pacific countries. They found IPOs to be more likely in countries with a higher legality index. [Lerner and Schoar \(2005\)](#)

² For traditional VCs (organized as limited partnerships), a VC firm refers to the VC fund management company while a VC fund refers to the entity that makes the investments. Since the detailed individual information of each fund manager is unavailable in the SDC VentureXpert database, we measure foreign VCs' human capital (including experience, networks and reputation) by VC firms instead of VC funds, which is the same as that in existing studies. This practice is acceptable since VC fund managers share their experience, network and reputation within their VC firms. And, although VC funds make investments, the strategic investment decisions are made by VC firms.

³ We separate the CEO from a top manager, where a top manager here means a top executive officer other than the CEO such as CFO, COO, CSO or CTO, etc.

investigated contract choices in 210 private equity transactions in emerging markets. They found that, unlike the U.S. VC market where convertible preferreds are ubiquitous, in emerging VC markets, a much broader array of securities are employed and private equity investors often have fewer contractual protections. The choice of securities appears to be driven by the legal and economic circumstances of the nation and the private equity group. Also, [Bruton et al. \(2004\)](#) examined how institutions in East Asia shape the VC industry and create differences from VCs in the West. [Ahlstrom and Bruton \(2006\)](#) explored how VCs in established markets use network relationships among entrepreneurs, VCs, enterprises and other informal institutions to manage investments in ways that supplement or replace weak, missing or unstable formal institutions in emerging markets. [Cumming et al. \(2008\)](#) examined the impact of VCs' ownership structures on VCs' provisions, especially in governance and value-added services, to small and medium-sized businesses (SMEs) in Japan. They found that U.S.-affiliated funds in Japan are more likely to have smaller portfolios and tend to provide more advice to their portfolio companies. [Cumming and Walz \(2010\)](#) further addressed the systematic biases in valuation reporting of private equity in 39 North and South American, European and Asian countries. Their findings imply that the reporting biases depend on international differences in accounting standards and laws, and on the proxy they used for information asymmetry between private equity managers and their institutional investors.

For the Chinese VC market, [Walker and Pukthuanthong \(2007\)](#) have shown that the Chinese VC market is different from established markets in terms of its cultural norms, corporate governance structures, exit strategies and government interventions. [Jingu and Kamiyama \(2008\)](#) have argued that the growth in China's VC market has been due to recent reforms. [Zeng \(2004\)](#) has presented a history of VC in China and suggests that the most restrictive policy in China is the difficulty involved in listing VC-backed companies on China's stock exchanges. [Fuller \(forthcoming\)](#) identifies three distinct patterns and proposes that several separate institutional arrangements are the main factors determining the behavior of VCs in China. Further, [Tan et al. \(2008\)](#) analyzed differences between foreign and domestic VCs in terms of the control and incentive mechanisms they use in China. They found that domestic VCs are less active in monitoring, less likely to retain veto rights and introduce stock options, and less motivated to provide value-added services than their foreign counterparts. [Lu et al. \(2007\)](#) further suggested that legal institutions contribute significantly to the performance of domestic and foreign VCs in China.

As in the U.S., the most prevalent exit method for VCs in Asia-Pacific markets is an IPO, which attracts many studies on the effects of VCs on IPOs. In the case of IPOs in Japan, [Hamao et al. \(2000\)](#) found that the long-run performance of VC-backed IPOs was more or less the same as that of other IPOs, with the exception of those firms backed by foreign-owned or independent VCs. However, they found that IPOs in which the lead VC firm was also the lead underwriter had higher first-day returns than other VC-backed IPOs. Their findings imply that a conflict of interest can influence initial pricing, but not the long-term performance of IPOs in Japan. [Da Silva Rosa et al. \(2003\)](#) explored initial underpricing and the long-run share performance of VC-backed IPOs in Australia. Their findings suggest no statistically significant differences in underpricing or long-run performance between VC-backed and other IPOs. [Sullivan and Unite \(2001\)](#) also concluded that VCs play a minimal role in supplying funds to companies that later go public in the Philippines. [Wang et al. \(2003\)](#) examined the effects of VCs on VC-backed listed companies in Singapore. They found the effects of the VCs' participation were very complicated; in particular, the post-IPO operating performance of VC-backed companies was inferior, although they were less underpriced. [Sun and Fang \(2009\)](#) investigate IPO underpricing and post-IPO operating performance of VC-backed Chinese firms. They examined the theories about VC certification, monitoring and reputation and they found that VC-backed IPOs exhibit larger underpricing and better post-IPO operating performance than non VC-backed IPOs. [Suchard \(2009\)](#) further investigated the role of VCs in corporate governance in Australian IPOs. He found that VC-backed IPOs are more likely to have an independent board, which is similar to the U.S. situation. VCs improve governance by using their networks to recruit more independent directors with industry experience.

In the second related literature, many studies have looked at the effect of the domestic environment on cross-border VC investments. [Alhorr et al. \(2008\)](#) suggest that economic integration policies influence the extent to which VCs invest in other member countries of the European Union. [Schertler and Tykvova \(2009\)](#) analyzed the impact of microeconomic and macroeconomic factors on the internationalization of venture capital. They identified a few factors, such as viable stock markets, that can boost investments from domestic and foreign VCs. [Aizenman and Kendall \(2008\)](#) have shown that distance, a common language,

and colonial ties are all significant factors in shaping cross-border VC and private equity flows. Schertler and Tykvova (2010) further studied the economic determinants of cross-border and domestic VC investments and found that patent counts, economic growth and stock markets have an important influence in attracting foreign VC investment. Balcarcel et al. (2008) analyzed a large sample of cross-border investments by U.S. VCs and found that the average size of an investment round and the fraction of funds raised in the first round are larger for companies in countries with poorer legal enforcement. Also, Cumming et al. (2009) have provided an analysis of international relocation decisions by VC-backed companies. They suggest that relocation of a VC-backed company to the U.S. is motivated by better economic conditions and the legal environment.

On foreign VCs' role in cross-border VC investments, Mäkelä and Maula (2005) addressed the role of foreign VCs in the internationalization of new ventures. They found that attracting investment from a VC firm in a venture's target market for internationalization can be valuable for the venture by helping to legitimize the unknown new venture in that market. Mäkelä and Maula (2006) further investigated interorganizational commitment in cross-border syndication networks. Their theory suggests that an investor's level of commitment, which is influenced by a portfolio company's prospects, is positively related to the investor's remoteness and negatively related to the relative investment size and the investor's embeddedness in local syndicate networks. Mäkelä and Maula (2008) explain the role of domestic VCs in attracting foreign VCs by pointing out that domestic VCs have several important roles in boosting cross-border investment, such as providing advice on operational management, signaling, and facilitating the formation of cross-border syndicates. Finally, recent work from Cumming et al. (2009) suggests that more experienced Asia-Pacific VCs have greater success with their investees' relocations to the U.S., and these relocations yield higher returns than staying in their countries of origin.

In the third related literature, studies of VC performance generally focus on domestic investment. They suggest that, besides financing, VCs improve performance through supplying a number of value-added services, particularly monitoring. The quality of VCs' monitoring appears to be recognized by the capital markets through smaller underpricing for IPOs with better monitoring (Barry et al., 1990) and better long-term performance (Brav and Gompers, 1997). VCs' human capital, such as experience, networks and reputation, is heavily emphasized in the literature. Gompers et al. (2008) investigated VCs' reactions to market signals during 1975–1998 and their efficiency consequences. They found that experience helps VCs to react correctly to market signals and increases the likelihood of success. Bottazzi et al. (2008) examined the determinants and performance consequences of European VCs' active monitoring and found that VCs with more experience will more actively monitor their portfolio companies. This active monitoring is shown to be positively related to the success of their portfolio companies. Also, Hochberg et al. (2007) studied the performance consequences of VCs' networks as measured by syndication relationships and concluded that better-networked VCs have significantly better performance. In a recent follow-up study, Hochberg et al. (2010) further confirmed the value of networks in restricting new entries of outside VCs, thus improving the incumbents' bargaining power with entrepreneurs. Nahata (2008) studied the impact of a VC firm's reputation on investment performance. He found that companies backed by more reputable VCs are more likely to exit successfully, access stock markets earlier, and have higher asset productivity at IPO.

Even though there is an expanding literature on Asia-Pacific VC markets and cross-border VC investments, there have been few studies focusing on the determinants of cross-border VC performance. We address this problem by examining the impact of foreign VCs' human capital, domestic entrepreneurs' experience and many other factors on cross-border VC performance in China. Our investigation finds no statistically significant relationship between foreign VCs' human capital and their cross-border performance in China. Instead, domestic entrepreneurs' characteristics were found to strongly affect VC performance. This finding is at variance with the results of some prior studies, but it is consistent with some recent studies on the impact of entrepreneurs. Gompers et al. (2010) suggest that an entrepreneur's successful experience helps her select the right industry and time to start a new venture. Chatterji (2009) has shown that in the medical device industry former employees of prominent companies tend to outperform other new entrants on a number of measures, including investment valuation, time in getting product approval, and time in obtaining the first funding. Liu and Yermack (2007) and also Malmendier and Tate (2005) have found such relationships between CEO characteristics and firm performance.

3. Data, variables and methodology

Our sample comes from the Zero2IPO database, which is the main data source for China's VC industry. To ensure data quality, we exclude VC investments made before 1999 and focus on VC investments made between 1999 and 2006 by foreign VCs in Chinese domestic companies. Since we trace the performance of portfolio companies until September 2009, the methodology provides a minimum of 2.5 years for a successful exit, where successful exit here means exit through an IPO or acquisition. Companies that had not exited successfully by September 2009 are classified as unsuccessful. Information on successful exits is available in the Zero2IPO database.

Non-random exit choices are a potential problem in our sample. Due to the limitations of the databases, we could not know the exact exit method of a portfolio company unless it was through an IPO or an acquisition. Like the SDC VentureXpert database, the Zero2IPO database discloses only IPOs and acquisitions for VC-backed companies. We have no detailed information on other routes of exits such as buyouts, secondary sales and write-offs. We thus could not separate non-exited investments from unsuccessfully exited investments in our sample. This makes it impossible for us to control for non-random exit decisions in a two-stage Heckman procedure as Cumming et al. (2006) and Cumming (2008) have done. To mitigate the potential problem of non-random exit choices, for each portfolio company in our sample during 1999–2006, we trace forward at least 2.5 years to identify an exit, bearing in mind that a foreign VC firm's operational period from its first investment disbursement to exit is much shorter in China, typically 2–3 years (Fang, 2006; Qian and Zhang, 2007). By this construction, we expect those VC investments that did not exit by September 2009 to make up only a very small proportion of our sample, which rules out non-random exit decisions as an alternative explanation for our findings. We thus classify our sample into two categories, one for successful exits (IPOs or acquisitions) and the other for unsuccessful exits (including liquidations, but also non-exits), even though there may only be a few non-exit observations in the latter category. This is similar to the approach taken by many previous researchers (Gompers and Lerner, 2000; Hochberg et al., 2007; Gompers et al., 2008; Nahata, 2008), who ignored the non-random exit choice problem by tracing forward at least 4 years for an exit considering the average investment period of around 5 years in U.S. VC investments (Cumming and MacIntosh, 2003).

To address foreign VCs' human capital from VC markets worldwide, we rely primarily on the SDC VentureXpert database from Thomson Financial. According to Kaplan et al. (2002), the SDC VentureXpert database captures around 85% of the financing rounds made by U.S. VCs. Du and Vertinsky (2008) point out that this database includes most of the venture-related VC deals in Europe and most of the big VC deals in Asia. Aizenman and Kendall (2008) also suggest that this database is a comprehensive data source for VC deals worldwide, with over 30 years of historical data. We trace each foreign VC firm's global investment information in this database and drop those investment observations by foreign VCs that have no information in the database. The resulting sample consists of 495 identifiable VC investments by 84 foreign VCs in 243 Chinese domestic companies for which relevant data on the portfolio companies and the VCs are available. We investigate both VC investments and the VC-backed portfolio companies. Among the 495 VC investments, each VC firm-portfolio company pair is unique, even though the VC firm may have participated in multiple rounds of financing.

Table 2 presents the distribution of our sample. The number of observations and the corresponding percentages (in parentheses) are listed. Panel A presents the distribution of VC investments by the funding year and the development stage of a portfolio company. As shown, foreign VC investments decreased during 2001–2002, presumably due to the impact of the internet bubble of the late 1990s. China's VC industry has since experienced rapid expansion during recent years. Most foreign VC investments are made at the seed/early and growth stages of development in portfolio companies, involving about 90% of VC investments, which implies high risks in these VC investments.

Panel B presents the distribution of portfolio companies by industry and Chinese region. The sample covers the following nine industries based on Zero2IPO and SDC VentureXpert's industry definitions⁴: biotechnology, communications, computer related, consumer related, industrial products, medical/health

⁴ SDC VentureXpert's industry classification is based on the VEIC (venture economics industry classification). Zero2IPO's industry classification is quite similar. We can match a VEIC code for each industry group in Zero2IPO, by which we classify industries in Zero2IPO into nine groups based on the VEIC codes. If we use Zero2IPO's industry classifications directly, all our results remain qualitatively the same.

Table 2

Sample distributions. The sample consists of 495 identifiable VC investments by 84 foreign VCs in 243 Chinese domestic companies that received their initial VC funding during 1999–2006 and for which data is available in the databases. The table presents the distribution of foreign VC investments by funding year and the development stage of portfolio companies in Panel A, the distribution of VC-backed portfolio companies by industry and Chinese region in Panel B, and the distribution of foreign VCs by global region and VC firm type in Panel C. Each panel presents the number of observations and the corresponding percentages (in parentheses).

Panel A: distribution of VC investments by funding year and development stage							
Development stage	Seed/early stage	Growth stage		Late stage	Total		
Funding year							
1999	16 (3.23)	10 (2.02)		5 (1.01)	31 (6.26)		
2000	20 (4.04)	24 (4.85)		0 (0)	44 (8.89)		
2001	9 (1.82)	8 (1.62)		1 (0.20)	18 (3.64)		
2002	7 (1.41)	14 (2.83)		0 (0)	21 (4.24)		
2003	10 (2.02)	22 (4.44)		8 (1.62)	40 (8.08)		
2004	20 (4.04)	54 (10.91)		5 (1.01)	79 (15.96)		
2005	21 (4.24)	77 (15.56)		15 (3.03)	113 (22.83)		
2006	48 (9.70)	88 (17.78)		13 (2.63)	149 (30.10)		
Total	151 (30.51)	297 (60.00)		47 (9.49)	495 (100)		
Panel B: distribution of portfolio companies by industry and Chinese region							
China region	East	Central	North	North-east	South	South-west	Total
Industry							
Biotechnology	1 (0.41)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	1 (0.41)
Communications	10 (4.12)	0 (0)	19 (7.82)	0 (0)	15 (6.17)	1 (0.41)	45 (18.52)
Computer related	39 (16.05)	1 (0.41)	59 (24.28)	1 (0.41)	12 (4.94)	2 (0.82)	114 (46.91)
Consumer related	4 (1.65)	0 (0)	2 (0.82)	0 (0)	1 (0.41)	0 (0)	7 (1.88)
Energy related	4 (1.65)	1 (0.41)	0 (0)	0 (0)	1 (0.41)	0 (0)	6 (2.47)
Industrial products	2 (0.82)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	2 (0.82)
Medical/health related	4 (1.65)	0 (0)	1 (0.41)	0 (0)	1 (0.41)	0 (0)	6 (2.47)
Other electronics related	18 (7.41)	1 (0.41)	5 (2.06)	0 (0)	7 (2.88)	0 (0)	31 (12.76)
Other services and manufacturing	15 (6.17)	1 (0.41)	11 (4.53)	0 (0)	4 (1.65)	0 (0)	31 (12.76)
Total	97 (39.92)	4 (1.65)	97 (39.92)	1 (0.41)	41 (16.87)	3 (1.23)	243 (100)
Panel C: distribution of VCs by global region and type							
Type	Corporate VC	Institutional VC	Government VC		Traditional VC	Total	
Global region							
Asia	6 (7.14)	2 (2.38)	0 (0)		14 (16.67)	22 (26.19)	
U.S.	5 (5.95)	6 (7.14)	2 (2.38)		46 (54.76)	59 (70.24)	
Europe	0 (0)	0 (0)	0 (0)		1 (1.19)	1 (1.19)	
Other	0 (0)	0 (0)	0 (0)		2 (2.38)	2 (2.38)	
Total	11 (13.10)	8 (9.52)	2 (2.38)		63 (75)	84 (100)	

related, electronics related, and other services and manufacturing. As in the U.S. VC market, the majority of Chinese portfolio companies belong to high-tech sectors, such as computer related, communications and electronics related, which constitute almost 80% of our sample. The industry distribution also suggests high risks in China's VC industry. We classify the locations of portfolio companies into six major regions in China, which are east, central, north, north-east, south, and south-west. As shown, most portfolio companies are located in east and north, each of which constitutes almost 40% each of our sample. Based on provinces (not reported), most of the portfolio companies are headquartered in Beijing, Shanghai and Guangdong provinces, which contribute roughly 40%, 31% and 14% to our sample, respectively. Only around 15% of the portfolio companies are located in other provinces of China.

Panel C shows the distribution of the VCs by global region and type, where the global regions are: the U. S., Asia, Europe, and others. As shown, most foreign VCs are from the U.S. and Asia, which constitute 70% and 26% of our sample, respectively. Following the literature on VC strategies (Bottazzi et al., 2008; Nahata, 2008; Hellmann et al., 2008), we identify four VC firm types: independent VCs, VCs affiliated with a

Table 3

Variable definitions, measures and sources.^a This table presents definitions, measures and data sources for the dependent, independent and control variables and some terminology. Indicator variables are indicated by *. Variables in natural logarithm are indicated by **.

Variables	Definition and measurement	Data source
Dependent variables		
Investment Success*	An indicator variable indicating whether the VC-backed portfolio company went public or was acquired before September 2009.	Zero2IPO
Investment Duration**	The logarithm of one plus the number of days from the date of VCs' first investment in the portfolio company to the date of exit. It is right-censored at September 2009 for those investments yet to exit successfully before September 2009.	Zero2IPO
Independent variables		
Foreign VCs' human capital		
VC Experience**	The logarithm of one plus a VC firm's age at the time of its first investment in the portfolio company.	VentureXpert
VC Network (%)	The normalized number of other VCs with whom the VC firm has syndicated in previous 5 years in VC markets worldwide at the time of its first investment in the portfolio company.	VentureXpert
VC Reputation (%)	The normalized market value of all companies taken public by the VC firm in VC markets worldwide from the beginning of 1994 until the time of its first investment in the portfolio company.	VentureXpert
Entrepreneurs' Experience		
CEO General Experience**	The logarithm of one plus the number of companies that the CEO has previously worked in whatever job at the time of her joining the portfolio company as CEO.	Zero2IPO
Special Experience as CEO**	The logarithm of one plus the number of companies that the CEO has previously served as CEO at the time of her joining the portfolio company as CEO.	Zero2IPO
Special Experience as Manager**	The logarithm of one plus the number of companies that the CEO has previously served as a top manager at the time of her joining the portfolio company as CEO.	Zero2IPO
Control variables		
Syndicate Size	The number of VCs who participated in the portfolio company's funding.	Zero2IPO
Portfolio Company Quality **	The logarithm of one plus the total VC funding amount across all financing rounds.	Zero2IPO
VC Industry Development**	The logarithm of one plus the lagged yearly aggregate VC fund inflows to China at the time of the portfolio company's receiving its initial VC funding.	Zero2IPO
IPO and M&A market conditions		
Domestic IPO Conditions**	The logarithm of one plus the lagged quarterly number of IPOs in China prior to the portfolio company's exit. For investments yet to exit successfully before September 2009, it is the logarithm of one plus the average of the lagged quarterly number of IPOs over the entire time period starting from the date of the portfolio company's receiving its initial VC funding.	CSMAR
Domestic M&A Conditions **	The logarithm of one plus the lagged quarterly number of M&A transactions in China prior to the portfolio company's exit. For VC investments that had not yet exited successfully before September 2009, it is the logarithm of one plus the average of the lagged quarterly number of M&A transactions over the entire time period starting from the date of the portfolio company's receiving its initial VC funding.	CSMAR
Chairman Overlap Indicator*	An indicator variable indicating whether the CEO of the portfolio company is also the chairman of the board at the time of her joining the portfolio company as CEO.	Zero2IPO
Founder Overlap Indicator*	An indicator variable indicating whether the CEO of the portfolio company is also the founder at the time of her joining the portfolio company as CEO.	Zero2IPO
Founder Departure Indicator*	An indicator variable indicating whether the founder has left the portfolio company when the CEO's joining the portfolio company as CEO.	Zero2IPO
Corporate VC Indicator*	An indicator variable indicating whether the VC firm is affiliated with a corporation.	VentureXpert
Institutional VC Indicator*	An indicator variable indicating whether the VC firm is affiliated with an institution.	VentureXpert
Early Investment Indicator*	An indicator variable indicating whether the first investment by a VC firm in the portfolio company occurred at the seed or early stage of development.	Zero2IPO
Misc.		
Local Availability of Experienced CEOs	A fraction, in which the denominator is the number of portfolio companies in our sample that headquartered in the same province as the specific portfolio company and the numerator is the number of portfolio companies that have hired CEOs who have previously worked for at least one other company whatever the job.	Zero2IPO
Portfolio Size	The number of companies in the VC firm's worldwide portfolio.	VentureXpert
Lead VC Firm	The VC firm that participated in the first round of financing and made the largest total investment in the company across all rounds.	Zero2IPO

corporation, VCs affiliated with financial institutions, and VCs affiliated with governments. For simplicity, we call them respectively traditional VCs, corporate VCs, institutional VCs and government VCs. As shown, 75% of the foreign VCs are traditional VCs. Corporate, institutional and government VCs constitute respectively 13.10%, 9.52% and 2.38% of our sample.

We present the definitions, measures and the sources of all the variables and some terminologies in Table 3. We define and explain the variables in the following subsections.

3.1. Performance measure

We intend to investigate the determinants of foreign VC performance in China. Ideally, we would measure performance directly, using the rate of return. However, such returns are not systematically available to the public, since VCs generally disclose their performance only to their investors. It is suggested that the majority of observable VC returns are from successful exits such as going public or being acquired (Cumming and MacIntosh, 2003; Cochrane, 2005). The more successful exits a VC firm has from its portfolio companies, the larger its internal rate of return from investment tends to be. Hence, we measure VC performance indirectly using the likelihood of a successful exit by September 2009, calling this variable *Investment Success*. A portfolio company is treated as successful if it goes public or is acquired. Similar measures have been used by Gompers and Lerner (2000), Brander et al. (2002), Hochberg et al. (2007), Sorensen (2007), Zarutskie (2007), and Nahata (2008).

3.2. Foreign VCs' human capital

Our aim is to analyze the roles of foreign VCs and domestic entrepreneurs in VC performance. In the literature, VCs' human capital is known as a major determinant of VC performance. Human capital in this context generally includes the experience, networks and reputation accumulated through experience in fundraising, project selection, investment and exit choices. In the case of cross-border VC investments in China, most foreign VCs have accumulated human capital in VC markets worldwide. We take foreign VCs' human capital as the primary explanatory variable in our model. At the VC investment level, this variable is measured for each VC firm at the time of its first investment in a portfolio company. At the portfolio company level, we take the lead VC firm's human capital as the primary explanatory variable at the same point in time. Typically, the lead VC firm initiates the first investment and is among the first group of investors for a portfolio company. The lead VC firm is often the most active investor and plays a leading role in monitoring and professionalizing the portfolio company. Following Nahata (2008), we define the lead VC firm as the firm that participated in the first round of financing and made the largest total investment in the company across all rounds. This identification of the lead VC firm is more appropriate than the one that classifies the lead VC firm as the one which made the largest cumulative investment across all financing rounds.

The following three variables are taken as proxies for foreign VCs' human capital. These proxies are conditional on the available information in SDC VentureXpert database.

- (1) *VC Experience*. Bottazzi et al. (2008) and Gompers et al. (2008) have shown a VC firm's experience to be an important determinant of its investment performance. The more experience a VC firm has, the more monitoring role it can apply to its portfolio companies, and the better the investment performance will be. Following Lee and Wahal (2004) and Nahata (2008), we use a foreign VC firm's age as a proxy for its experience. Information on each foreign VC firm's age is available in both the Zero2IPO and SDC VentureXpert databases. We further double check this information against each VC firm's website. We also use the cumulative number of a VC firm's investment rounds in all its investments as an alternative proxy for *VC Experience* (Sorensen, 2007; Nahata, 2008), and the results are qualitatively similar.
- (2) *VC Networks*. Hochberg et al. (2007) investigated the impact of a VC firm's syndication networks on its investment performance. They found a positive relationship between networks and performance.

Notes to Table 3:

^a We present the measures of all the variables in terms of VC investments and the corresponding foreign VCs and domestic CEOs. For the analysis at the portfolio company level, we need to measure them in terms of portfolio companies and the corresponding foreign lead VCs and domestic CEOs at the time a portfolio company first receives VC funding.

The explanation is that VCs with better networks have better access to future deal flows, investment opportunities and information. Hochberg et al. (2010) further concluded that *VC Networks* can be used by current VCs to prevent new entry. The existing VCs benefit from reduced entry by demanding better deals for their investments. Following Hochberg et al. (2007) and Nahata (2008), we measure a VC firm's networks in terms of the number of other VCs it had syndicated within the previous 5 years on a rolling basis and normalize it by the number of active VCs participating in at least one investment deal during the 5-year span. For example, for a VC firm *i*, if it syndicated with 4 different VCs between 2000 and 2004 and there were a total of 21 active VCs participating in at least one investment deal during this period, then the value of networks for firm *i* in the year 2005 would be $20\% (4/(21-1))$.

- (3) *VC Reputation*. Nahata (2008) has shown that the more reputable a VC firm is, the more likely its portfolio companies will have a successful exit (by IPO or acquisition). Consistent with this, we measure a foreign VC firm's reputation using the cumulative market capitalization of the IPOs it had backed worldwide. For each VC firm, we sum the market value of all the portfolio companies it had taken public from the beginning of year 1994 until the given year and normalize it by the aggregate market value of all VC-backed companies that went public from the beginning of 1994 to the same year. The market value of a portfolio company is based on the closing share price and the shares outstanding on the IPO day. Such information is available in SDC VentureXpert database. Since the lead VC firm's reputation is measured at the time of the portfolio company's receiving its first VC funding, the measure incorporates at least 5 years of VC performance if the portfolio company was initially funded in 1999 (in this case, the lead VC firm's reputation will be based on those companies taken public by this VC firm between 1994 and 1998). For companies initially funded in 2005, this reputation measure incorporates 11 years of VC performance from 1994 to 2004. For example, if the SDC VentureXpert database shows that VC firm *i* backed 3 companies that went public between 1994 and 2000 with a combined market value of 15 million US dollars, and during the same period a total of 20 VC-backed companies went public with a total market value of 300 million US dollars, then the reputation value of VC firm *i* in year 2001 is $5\% (15/300)$.

3.3. Domestic entrepreneurs' experience

In the existing literature on VC, entrepreneurs have received little attention due to data limitations, even though they may play a key role in VC performance. The Zero2IPO database provides detailed information on top management, especially on the CEO of each VC-backed company, including his/her name, home address, position, education and working experience. Such information allows us to analyze the impact of domestic entrepreneurs on VC performance. We hand-collect information on the CEOs from the descriptions of individual managers in the Zero2IPO database.

As a key factor on domestic entrepreneurs' human capital, experience is taken as a primary explanatory variable in our analysis. We construct three variables as proxies for CEO experience. The first is the number of companies in which a CEO had previously been employed in any capacity. The second is the number of companies of which the CEO had previously served as CEO. And, the third is the number of companies in which the CEO had previously served as a top manager, such as CFO, COO, CSO or CTO. The first variable indicates a CEO's general experience, while the second and third variables indicate a CEO's special experience as CEO or top manager.

At the VC investment level, these variables are measured for each CEO at the time of her joining the company as CEO. At the portfolio company level, we take these factors of the CEO at the time of the portfolio company's receiving its first VC funding as the primary explanatory variables. This treatment also applies to foreign VCs' human capital.

3.4. Control variables

Following the literature and taking into account characteristics of the Chinese market, we further control for domestic entrepreneurs' other characteristics, VC firm type, VC syndication, portfolio company characteristics, domestic *VC Industry Development*, domestic market conditions for IPO and M&A, and indicators accounting for fixed effects. We now explain these variables.

3.4.1. Domestic entrepreneurs' other characteristics

In China, an overlap of the CEO and the chairman of the board is common and the largest shareholder typically holds more than half of the equity. The effect of this overlap is largely unknown, especially for nominally private companies. To rule out the possible impact of this overlap on *Investment Success*, we control for the *Chairman Overlap Indicator* indicating whether the CEO is also the chairman.

In our sample, a considerable proportion of the VC-backed companies are family companies. Hence, having a separate CEO and founder may indicate the company's transition to a modern corporation. Hence, we control for the *Founder Overlap Indicator* indicating whether the CEO is also the founder of the portfolio company. To take into account possible influence of the founder on the CEO, we further control for the *Founder Departure Indicator* indicating whether the founder has left the company. Due to data limitations, we cannot deduce the exact time of when the founder had left a company if indeed he had. Hence, we treat the founder as having left if he held no formal position in the top management team and did not sit on the board of directors.

Similar to the earlier primary explanatory variables, these factors are measured at the time of the CEO's joining the portfolio company as CEO in our analyses of VC investments, and they are measured at the time of the portfolio company's receiving its first VC funding in our analysis of portfolio companies.

3.4.2. VC firm type

In terms of organizational structure, there is a distinction between traditional VCs and so-called captive VCs. The latter are owned by corporations, financial institutions or governments (Bottazzi et al., 2008; Nahata, 2008; Hellmann et al., 2008; Du and Vertinsky, 2008). While traditional VCs aim to maximize the value of a limited partnership, captive VCs may emphasize certain strategic objectives of their parents, sacrificing at times opportunities to maximize the value of their investments. In order to control for any impact of this difference in objectives on *Investment Success*, we create two indicator variables indicating whether a VC firm is a corporate or an institutional VC firm. Since government VCs constitute only a very small proportion of our sample, there is no need to control for government VCs.

3.4.3. VC syndication

Casamatta and Haritchabalet (2007) have shown theoretically that syndication improves the screening of VC deals and prevents competition among investors after investment opportunities are disclosed. Brander et al. (2002) have provided evidence that syndicated VC investments have higher returns, using a sample of Canadian companies. Hence, we control for the size of the VC syndicate in each portfolio company.

3.4.4. Portfolio company characteristics

Better quality companies are more likely to be associated with better entrepreneurs and better VCs. A portfolio company's general quality is, by definition, likely to have an impact on performance. Hence, we control for portfolio company quality.

Greater total VC investment is likely to be associated with better portfolio company quality, and thus with a greater likelihood of a successful exit. Gompers (1995) proposed that funding in follow-on rounds occurs only if a portfolio company does well in earlier rounds. Also, specific milestone-contingent clauses are contained in most VC financing contracts by which VCs automatically stop providing additional rounds of financing to weak ventures (Wang and Wang, 2009). Mäkelä and Maula (2006) have shown that changes in a portfolio company's prospects influence investors' commitment levels in a model of cross-border VC investments. So greater VC funding reflects better quality in a portfolio company. Following Nahata (2008), we use the total VC investment in a portfolio company across all financing rounds as a proxy for its quality.

Of course, this measure of company quality is imperfect and involves a look-ahead bias since the cumulative funding for a portfolio company is unknown at the time of earlier financing rounds (Nahata, 2008). However, we are unable to devise a more suitable quality measure, since the Zero2IPO and VentureXpert databases provide little information on portfolio companies other than the number of financing rounds, the funding amount, the investors, and exits. As a robustness check, we exclude this company quality variable and re-run all the regressions. All the results remain qualitatively unchanged.

We also control for the *Early Investment Indicator* indicating whether the first VC investment in a portfolio company occurs at the seed or early stage of development. Companies in the early stages of development are likely to be riskier and this may affect VC performance.

3.4.5. Domestic VC Industry Development

China's VC industry is growing rapidly. It is reasonable to expect a positive linkage between *VC Industry Development* and individual VC performance. Hence, we control for China's *VC Industry Development* in our regressions using the lagged aggregate funds inflow into China's VC industry in the year prior to a VC firm's first investment in a portfolio company.⁵

3.4.6. Domestic market conditions for IPO and M&A

Lerner (1994) suggests that VCs choose a good time to exit, particularly in the case of an IPO. Hence, the timing of a VC firm's exit from a portfolio company through an IPO or acquisition is likely to depend on conditions in the IPO and M&A markets. We thus control for the exit market environment using two variables similar to those used by Hochberg et al. (2007) and Nahata (2008), each measuring the state of its respective market. Specifically, we use the quarterly numbers of IPOs and M&As respectively over a period of 3 months prior to a portfolio company's exit. These two measures are lagged by a quarter to allow a period for a company and its investors to prepare for an impending exit. Since information on the number of M&As in private markets is unavailable in China, we use the number of M&As in public markets as a proxy for the whole market, assuming a high correlation between public and private markets, where public and private markets consist of reporting and non-reporting companies, respectively. We use data for these two variables from the CSMAR database, which contains information on China's stock markets and public companies.

For unsuccessful exits, it is not obvious how to measure exit market conditions. Following Nahata (2008), we use the average of the lagged quarterly number of IPOs and M&A transactions over the entire time period starting from the date of a portfolio company's receiving its first VC funding as a proxy for unsuccessful exit. In the Cox hazard model, we allow for time-varying exit market conditions, which would alleviate the problem of measuring unsuccessful exits.

3.4.7. Other factors

We include as well industry indicators to partially account for technological and industrial characteristics of portfolio companies. We also include year fixed effects, based on initial VC funding to a portfolio company, as additional control variables. Considering the possible effect of earlier success on later rounds, we further control for round sequence fixed effects in the analysis of VC investments.

3.5. Methodology

We use logit and Cox hazard models to analyze VC performance at both the VC investment level and the portfolio company level. At the VC investment level, we have one observation for each pair of foreign VC firm and portfolio company. At the portfolio company level, we have one observation for each portfolio company. The dependent variable of the logit model is *Investment Success*, which takes the value of 1 if a VC-backed portfolio company went public or was acquired before September 2009, and 0 otherwise. The dependent variable of the Cox hazard model is *Investment Duration*,⁶ which is the logarithm of the duration from the date of a VC firm's first investment in a portfolio company to the date of exit. It is right-censored at September 2009 for those companies that had not yet exited successfully by September 2009. The Cox hazard model allows right-censored data and is a semi-parametric model in which the hazard function is not dependent on a specific distribution of the survival time.

⁵ This variable is often used as a proxy for VC industry competition in developed VC markets (Gompers, 1995; Gompers and Lerner, 2000; Hochberg et al., 2007; Nahata, 2008). But in emerging markets such as China, this variable is more suitable as a proxy for *VC Industry Development*.

⁶ Giot and Schwienbacher (2007) and Cumming and Johan (2010) focused specifically on VC investment duration. Giot and Schwienbacher (2007) analyzed the time to IPO, trade sales and liquidations for 6000 U.S. VC-backed firms. They found that the hazard rate for an IPO is clearly non-monotonic with respect to time. They also provided evidence of the impact of economic factors such as syndicate size, composition, geographic location and VC value added on the exit outcome. Cumming and Johan (2010) formulated a theory of VC investment duration based on the idea that a VC exits when the expected marginal cost of maintaining the investment exceeds the expected marginal benefit, and they thereby related VC investment durations to portfolio company characteristics, the VCs' characteristics, the deal's characteristics, and institutional and market conditions. We control for most of these factors known to influence VC investment duration.

All our primary explanatory variables, including foreign VCs' human capital and domestic entrepreneurs' experience, are measured at the time of either the VC firm's first investment in a portfolio company or the CEO's joining the portfolio company as CEO. Specifically, at the investment level, foreign VCs' human capital is measured for each VC firm at the time of its first investment in the portfolio company, and domestic entrepreneurs' experience is measured for each CEO at the time of her joining the portfolio company as CEO. At the portfolio company level, these factors for the lead VC firm and the CEO, measured at the time of the portfolio company's receiving its first VC funding, are used as the primary explanatory variables. Similarly, domestic entrepreneurs' other characteristics, including the *Chairman Overlap Indicator*, *Founder Overlap Indicator* and *Founder Departure Indicator*, are also measured at the time of the CEO's joining the portfolio company as CEO. *VC Industry Development*, as proxied by the aggregate VC fund inflows into the Chinese VC industry, is also measured at the time of each VC firm's first VC investment in a portfolio company.

There is typically a substantial time gap between a foreign VC firm's first investment and its exit, as well as between a CEO's joining a portfolio company as CEO and the company's IPO or acquisition. We relate foreign VCs' human capital and CEO experience accumulated in the past to future VC performance. This methodology can effectively address concerns about reverse causality whereby better performance might lead to better CEO experience and better VCs' human capital. Similar methodologies were used by Hochberg et al. (2007) and Nahata (2008).

3.6. Summary statistics

Table 4 presents summary statistics at the portfolio company and VC investment levels in Panels A and B, respectively.⁷ The quartiles, means, standard deviations and the number of observations are presented. We have very similar data distributions in Panels A and B.

As shown, 36% of the portfolio companies exited successfully through IPOs or acquisitions, which is higher than the 25% reported by Nahata (2008) and the 26% reported by Hochberg et al. (2007) for the U.S. domestic VC industry. The average *Investment Duration* of VC investments in China is 1433 days (almost 4 years),⁸ which is substantially shorter than the 5.5 years reported by Nahata (2008). The average age, networks and reputation of the lead VCs are roughly 22 years old, 3.24% and 1.22%, respectively, at the time of a portfolio company's receiving initial VC funding. Compared with that of the U.S. counterpart reported by Nahata (2008), the foreign VCs in our sample generally have more experience, less extensive networks but comparable reputations. As for the domestic entrepreneurs' experience, the CEO at the time of a portfolio company's receiving initial VC funding has on average worked in 2 other companies, served as CEO of 0.4 other companies, and as a top manager of 1 other company.

Roughly 33% and 28% of the CEOs are respectively also the chairman of the board and the founder of their company. Since most of the VC-backed companies are private enterprises, often family enterprises, an overlap of CEO and chairman or founder is much more common than in developed countries. As shown, 34% of founders have left the companies. Around 12% and 9% of lead VCs are respectively corporate and institutional VCs, which is comparable with the percentages reported by Bottazzi et al. (2008) and Nahata (2008) for European and U.S. VCs, respectively.

On average, three VCs form a syndicate in a portfolio company, which is slightly lower than that in the U.S. VC industry as reported in Lerner (1994), Kaplan and Strömberg (2003) and Nahata (2008). The total VC funding across all financing rounds in a portfolio company is 38.08 million US dollars on average, which is comparable with that reported by Nahata (2008). Around 31% of portfolio companies receive their initial VC funding at an early/seed development stage, which is slightly lower than that reported by Nahata (2008) for the U.S. VC industry during 1991–2001. The aggregate VC fund inflows in the year prior to a VC firm's first investment in a portfolio company is around 875.61 million US dollars, which is much lower than that in the U.S. VC market as reported by Gompers (1995), Hochberg et al. (2007) and Nahata (2008), suggesting substantial under-development in China's VC industry. Also, there are on average 15.5 IPOs and

⁷ We report the original values for all the variables in Table 4, even though some of them are taken logarithms in our regressions (as indicated by "****" in Table 3).

⁸ The average *Investment Duration* is 966 days for successful exits and 1433 days for all VC investments (taking into account the fact that we do not know the exact time of an unsuccessful exit and unsuccessful exits are right-censored at September 2009).

Table 4

Summary statistics for VC-backed portfolio companies and VC investments during 1999–2006. Panel A presents statistics describing VC-backed portfolio companies initially funded by foreign VCs between 1999 and 2006 and which are covered by the databases. We trace the performance of the sample until September 2009, which allows at least 2.5 years for an exit. The sample is classified into two categories, one for successful exits (IPOs or acquisitions) and the other for unsuccessful exits (liquidations, including non-exits). *Investment Success* takes the value of 1 if the portfolio company had gone public or had been acquired before September 2009 and 0 otherwise. *Investment Duration* of a successful portfolio company that had either gone public or been acquired is the calendar time from the date of its receiving initial VC funding to the date of exit. *Investment Duration* of portfolio companies that had not exited successfully by September 2009 is right-censored at September 2009. The foreign lead VC firm's human capital is represented by three variables: *VC Experience*, *VC Networks*, and *VC Reputation*. Domestic entrepreneurs' experience is represented by three variables: *CEO General Experience*, *Special Experience as CEO*, and *Special Experience as Manager*. Other variables include *Syndicate Size*, *Portfolio Company Quality*, *VC Industry Development*, *Domestic IPO Conditions*, *Domestic M&A Conditions*, *Chairman Overlap Indicator*, *Founder Overlap Indicator*, *Founder Departure Indicator*, *Corporate VC Indicator*, *Institutional VC Indicator*, and *Early Investment Indicator*. The definitions, measures and data sources for the variables are described in Table 3. Panel B presents similar statistics for foreign VC investments between 1999 and 2006 in Chinese domestic companies that received initial VC funding during 1999–2006 and have relevant data in the databases. The quartiles, means, standard deviations and the number of observations are presented. As mentioned at the start of Section 3, because the average *Investment Duration* of foreign VC investments in China from their first investment to exit is typically 2–3 years (Fang, 2006; Qian and Zhang, 2007), by tracing at least 2.5 years for an exit, we expect those observations from companies that did not exit by September 2009 to make up only a very small proportion of our sample. Hence, we classify the sample of non-exit investments in September 2009, if any, as liquidations in this and all subsequent tables.

	0.25	0.5	0.75	Mean	Std.	Obs.
<i>Panel A: portfolio company level</i>						
Investment Success				0.36	0.48	243
Investment Duration (days)	1021	1356	1874	1433.29	799.61	243
Foreign lead VCs' human capital						
VC Experience (years)	5	11	17	22.01	33.72	243
VC Networks (%)	0.48	1.92	3.05	3.24	4.58	243
VC Reputation (%)	0.02	0.44	1.3	1.22	1.95	243
Domestic entrepreneurs' experience						
CEO General Experience	0	2	3	2.01	1.87	243
Special Experience as CEO	0	0	1	0.4	0.79	243
Special Experience as Manager	0	1	2	0.99	1.25	243
Syndicate Size	1	2	4	2.83	2	243
Portfolio Company Quality (US\$M)	3	11.5	36	38.08	120.47	243
VC Industry Development (US\$M)	361.03	696.82	2098.54	975.17	775.17	243
IPO and M&A market conditions						
Domestic IPO Conditions	3	14	21	15.51	13.4	243
Domestic M&A Conditions	84	206	503	317.24	262.66	243
Chairman Overlap Indicator	0	0	1	0.33	0.47	243
Founder Overlap Indicator				0.28	0.45	243
Founder Departure Indicator				0.34	0.48	243
Corporate VC Indicator	0	0	0	0.12	0.33	243
Institutional VC Indicator	0	0	0	0.09	0.28	243
Early Investment Indicator	0	0	1	0.41	0.49	243
<i>Panel B: VC investment level</i>						
Investment Success				0.38	0.49	495
Investment Duration (days)	1021	1417	1874	1468.13	754.36	495
Foreign VCs' human capital						
VC Experience (years)	4	10	17	20.65	32.18	495
VC Networks (%)	0.46	2.03	3.73	3.40	4.73	495
VC Reputation (%)	0	0.16	1.30	1.15	1.91	495
Domestic entrepreneurs' experience						
CEO General Experience	1	2	3	2.02	1.80	495
Special Experience as CEO	0	0	1	0.43	0.83	495
Special Experience as Manager	0	1	2	1.07	1.32	495
Syndicate Size	2	4	5	4.26	3.12	495
Portfolio Company Quality (US\$M)	6	22.6	55.1	86.90	275.56	495
VC Industry Development (US\$M)	361.03	689.72	1117.96	875.61	731.47	495
IPO and M&A market conditions						
Domestic IPO Conditions	3	14	21	15.09	13.16	495
Domestic M&A Conditions	96	206	503	328.24	266.41	495

Table 4 (continued)

	0.25	0.5	0.75	Mean	Std.	Obs.
Chairman Overlap Indicator				0.35	0.48	495
Founder Overlap Indicator				0.29	.45	495
Founder Departure Indicator				0.37	0.48	495
Corporate VC Indicator				0.12	0.33	495
Institutional VC Indicator				0.08	0.28	495
Early Investment Indicator				0.31	0.46	495

317 M&As in the quarter prior to a portfolio company's exit, which is much lower than that in the U.S., suggesting substantial under-development in China's public markets.

4. Analysis at the VC investment level

In this section, we investigate the impact of foreign VCs' human capital and domestic entrepreneurs' experience on the performance of VC investments. The unit of analysis is VC investment, i.e., our sample consists of observation pairs of a foreign VC firm and a portfolio company. The sample size is 495. We take foreign VCs' human capital and domestic CEO experience for each VC investment as our primary explanatory variables.

We first present a logit analysis with *Investment Success* as the dependent variable. *Investment Success* takes the value of 1 if a VC-backed portfolio company had gone public or had been acquired by September 2009 and 0 otherwise. We investigate the determinants of the performance of VC investment in a logit model by relating *Investment Success* to the following primary explanatory variables: foreign VCs' human capital, including *VC Experience*, *VC Networks* and *VC Reputation*, and domestic entrepreneurs' experience, including *CEO General Experience*, *Special Experience as CEO* and *Special Experience as Manager*. Other control variables include *Syndicate Size*, *Portfolio Company Quality*, *VC Industry Development*, *Domestic IPO Conditions*, *Domestic M&A Conditions*, and indicators including the *Chairman Overlap Indicator*, *Founder Overlap Indicator*, *Founder Departure Indicator*, *Corporate VC Indicator*, *Institutional VC Indicator*, and *Early Investment Indicator*. The definitions, measures and data sources of all these variables are listed in Table 3. Although not reported, as additional control variables, we also include industry fixed effects based on 9 industries, year fixed effects based on the initial VC funding in a portfolio company, and round sequence fixed effects based on the sequencing of investment rounds in each portfolio company (being in the first, second or third round, etc.).

Table 5 reports the results. All *t*-statistics in parentheses have been adjusted for heteroskedasticity (White, 1980) and clustered by company and year (Petersen, 2009) to account for correlations within a company and in a year. In models 1 through 4, we separately include foreign VCs' human capital and the three CEO experience variables. None of the three variables representing foreign VCs' human capital (experience, networks and reputation) is found to be significant. For example, in model 2, the coefficients of foreign VCs' experience, networks and reputation are respectively -0.08 , -0.08 and 0.06 , with *t*-statistics of -1.11 , -0.73 and 0.32 , none of which is statistically significant. That is, VCs' human capital, which is shown to be essential in the literature to VC performance in the U.S. market, plays little role in their cross-border investment performance in China. Our explanation to this is that foreign VCs' human capital from other markets is market-specific: such human capital has little value in an emerging market, such as the Chinese market.

In contrast, we find a significant impact for domestic entrepreneurs' experience. That is, the CEO's past experience has a significant positive effect on *Investment Success*. The coefficients of *CEO General Experience*, *Special Experience as CEO* and *Special Experience as Manager* are all quantitatively large with significant *t*-statistics. Specifically, a one standard deviation increase in *CEO General Experience*, *Special Experience as CEO* or *Special Experience as Manager* increases the odds of *Investment Success* by 30.9%, 31.3% and 22.3%, respectively.

Some other control variables may also affect the performance of VC investments. As expected, we find that *Portfolio Company Quality* positively affects VC performance. The better the quality, the greater the VC investment amount and also the higher the likelihood of a successful exit. Also, *VC Industry Development* positively affects VC performance. Better *VC Industry Development* signals good news, which may increase

Table 5

Logit analysis of *Investment Success* at the VC investment level. The sample consists of investments made between 1999 and 2006 by foreign VCs in Chinese domestic companies that received their initial VC funding during 1999–2006 and that have relevant data in the databases. We trace the performance of the sample until September 2009, which allows at least 2.5 years for an exit. The sample is classified into two categories, one for successful exits (IPOs or acquisitions) and the other for unsuccessful exits (liquidations, including non-exits). The dependent variable is *Investment Success*, which takes the value of 1 if the VC investment's portfolio company had a successful exit before September 2009 and 0 otherwise. Foreign VCs' human capital includes three variables: *VC Experience*, *VC Networks*, and *VC Reputation*. Domestic entrepreneurs' experience includes three variables: *CEO General Experience*, *Special Experience as CEO*, and *Special Experience as Manager*. Other variables include *Syndicate Size*, *Portfolio Company Quality*, *VC Industry Development*, *Domestic IPO Conditions*, *Domestic M&A Conditions*, *Chairman Overlap Indicator*, *Founder Overlap Indicator*, *Founder Departure Indicator*, *Corporate VC Indicator*, *Institutional VC Indicator*, and *Early Investment Indicator*. The definitions, measures and data sources of all variables are described in Table 3. Industry, initial funding year and round sequence fixed effects are also included in all regressions (not reported). Intercepts are not reported. Robust t-values, cluster-adjusted by portfolio company and year, are in parentheses. The significance levels at 1%, 5% and 10% are identified by ***, ** and *, respectively.

	1		2		3		4	
	Coeff.	t-Value	Coeff.	t-Value	Coeff.	t-Value	Coeff.	t-Value
Foreign VCs' human capital								
VC Experience	−0.09	(−1.43)	−0.08	(−1.11)	−0.15	(−1.51)	−0.08	(−1.22)
VC Networks	−0.08	(−0.78)	−0.08	(−0.73)	−0.08	(−0.71)	−0.08	(−0.73)
VC Reputation	0.06	(0.32)	0.06	(0.32)	0.07	(0.35)	0.05	(0.29)
Domestic entrepreneurs' experience								
CEO General Experience			0.45	(3.37)***				
Special Experience as CEO					0.64	(2)**		
Special Experience as Manager							0.35	(3.33)***
Syndicate Size	−0.02	(−0.29)	−0.03	(−0.56)	−0.05	(−1.38)	−0.04	(−0.72)
Portfolio Company Quality	0.30	(5.33)***	0.29	(5.7)***	0.27	(4.09)***	0.30	(5.51)***
VC Industry Development	2.01	(14.35)***	2.08	(13.26)***	2.11	(13.04)***	2.07	(14.99)***
IPO and M&A market conditions								
Domestic IPO Conditions	0.05	(0.26)	0.03	(0.19)	0.07	(0.42)	0.05	(0.26)
Domestic M&A Conditions	0.16	(0.73)	0.17	(0.75)	0.20	(0.88)	0.20	(0.88)
Chairman Overlap Indicator	0.54	(1.61)	0.55	(0.83)	0.65	(2.94)***	0.60	(1.9)*
Founder Overlap Indicator	−2.71	(−2.18)**	−2.69	(−2.24)**	−2.75	(−2.26)**	−2.65	(−2.13)**
Founder Departure Indicator	2.37	(2.88)***	2.29	(2.94)***	2.38	(2.88)***	2.35	(2.79)***
Corporate VC Indicator	1.16	(1.01)	1.05	(1.07)	1.09	(1.12)	1.07	(1.14)
Institutional VC Indicator	0.64	(2.25)**	0.64	(2.11)**	0.69	(2.04)**	0.66	(2.25)**
Early Investment Indicator	0.13	(0.7)	0.19	(0.83)	0.23	(0.83)	0.18	(0.82)
Industry, year and round sequence fixed effects	Present		Present		Present		Present	
Log likelihood	216		214		214		215	
Number of observations	486		486		486		486	
Pseudo-R ²	0.33		0.34		0.34		0.34	

the likelihood of a successful exit. Also, the *Founder Departure Indicator* and *Founder Overlap Indicator* respectively affect *Investment Success* positively and negatively. These results make sense. A departure of the founder and a separation of CEO and founder may imply a transition from a family enterprise to a modern corporation. Such a successful transition is expected to have a positive effect on performance. Further, the *Institutional VC Indicator* has a positive effect on *Investment Success*. Investments by institutional VCs are more likely to do well, probably due to their valuable services to portfolio companies.

The coefficients of the variables describing domestic market conditions for IPOs and M&As are almost insignificant, suggesting that cross-border VC performance is not very dependent on domestic conditions for IPOs and M&As. This is quite different from the findings in the literature for the U.S. VC market, as reported by Lerner (1994), Hochberg et al. (2007) and Nahata (2008). One possible reason is that, during our sample period 1999–2006, many foreign VCs exit via IPOs and M&As in overseas markets instead of in the domestic market. As indicated by Zeng (2004), the most restrictive policy in China is the difficulty to list a company on China's stock exchanges. We expect a substantial improvement of this situation over time with the development of China's capital market. This situation is indeed improving. The improvement of the Middle-and-Small Enterprise Board and the recent establishment of the Grown Enterprise Board are indications of China going in the right direction. Also, as indicated by Zero2IPO, the number of VC exits through domestic IPOs during 2006–2008 has reached 70, which constitutes 28% of all IPOs.

Other factors are insignificant, including *Syndicate Size*, *Chairman Overlap Indicator*, *Corporate VC Indicator*, and *Early Investment Indicator*.

We next turn to analyzing VC investments using a Cox hazard model. The dependent variable is *Investment Duration*, which for each successful VC investment is the logarithm of one plus the number of days from the date of the first investment in a portfolio company to the exit date. It is right-censored at September 2009 for those VC investments that had not exited successfully by September 2009. Table 6 reports the results. In the Cox hazard model a positive (negative) coefficient of a variable implies higher (lower) hazard for that variable and hence a shorter (longer) expected *Investment Duration*. Most of the results mirror the evidence reported earlier. In particular, the coefficients for foreign VCs' human capital are all insignificant, while the coefficients of *CEO General Experience*, *Special Experience as CEO* and *Special Experience as Manager* are highly significant. Specifically, a one standard deviation increase in *CEO General Experience*, *Special Experience as CEO* or *Special Experience as Manager* is associated with a 27.9%, 22.1% or 18.8% increase in the hazard of a VC firm's successful exit, respectively.

Table 6

Cox hazard analysis of *Investment Duration* at the VC investment level. The sample consists of investments between 1999 and 2006 by foreign VCs in Chinese domestic companies that received their initial VC funding during 1999–2006 and that have relevant data in the databases. We trace the performance of the sample until September 2009, which allows at least 2.5 years for an exit. The sample is classified into two categories, one for successful exits (IPOs or acquisitions) and the other for unsuccessful exits (liquidations, including non-exits). The Cox hazard model is used and the dependent variable is *Investment Duration*. *Investment Duration* of successful VC investments (which invested in portfolio companies that either went public or were acquired) is the logarithm of one plus the number of days from the date of the VC firm's first investment in a portfolio company to the date of exit. It is right-censored at September 2009 for VC investments that had not exited successfully by September 2009. A positive (negative) coefficient in the Cox hazard model indicates that the covariate increases (decreases) the hazard and shortens (lengthens) the expected duration. Foreign VCs' human capital includes three variables: *VC Experience*, *VC Networks*, and *VC Reputation*. Domestic entrepreneurs' experience includes three variables: *CEO General Experience*, *Special Experience as CEO*, and *Special Experience as Manager*. Other variables include *Syndicate Size*, *Portfolio Company Quality*, *VC Industry Development*, *Domestic IPO Conditions*, *Domestic M&A Conditions*, *Chairman Overlap Indicator*, *Founder Overlap Indicator*, *Founder Departure Indicator*, *Corporate VC Indicator*, *Institutional VC Indicator*, and *Early Investment Indicator*. The definitions, measures and data sources of the variables are described in Table 3. Industry, initial funding year and round sequence fixed effects are included in all regressions (not reported). Intercepts are not reported. Robust *t*-values, cluster-adjusted by portfolio company and year, are in parentheses. The significance levels at 1%, 5% and 10% are identified by ***, ** and *, respectively.

	1		2		3		4	
	Coeff.	t-Value	Coeff.	t-Value	Coeff.	t-Value	Coeff.	t-Value
Foreign VCs' human capital								
VC Experience	−0.11	(−1.09)	−0.11	(−0.63)	−0.11	(−0.75)	−0.11	(−0.81)
VC Networks	−0.02	(−0.46)	−0.02	(−0.38)	−0.02	(−0.37)	−0.02	(−0.41)
VC Reputation	−0.03	(−0.48)	−0.04	(−0.59)	−0.02	(−0.37)	−0.03	(−0.62)
Domestic entrepreneurs' experience								
CEO General Experience			0.41	(3.35)***				
Special Experience as CEO					0.47	(1.91)**		
Special Experience as Manager							0.30	(3.96)***
Syndicate Size	0.04	(1.14)	0.02	(0.89)	0.02	(1.07)	0.02	(0.75)
Portfolio Company Quality	0.20	(3.56)***	0.19	(4.45)***	0.18	(5.23)***	0.20	(3.45)***
VC Industry Development	0.50	(2.36)**	0.52	(2.1)**	0.56	(2.67)***	0.52	(2.46)**
IPO and M&A market conditions								
Domestic IPO Conditions	0.00	(0.06)	−0.01	(−0.15)	0.01	(0.26)	0.00	(−0.06)
Domestic M&A Conditions	−0.05	(−0.27)	−0.06	(−0.31)	−0.05	(−0.23)	−0.03	(−0.17)
Chairman Overlap Indicator	0.36	(1.57)	0.34	(1.65)*	0.46	(2.25)**	0.42	(2.05)**
Founder Overlap Indicator	−1.29	(−3.14)***	−1.31	(−3.28)***	−1.23	(−3.47)***	−1.22	(−3.22)***
Founder Departure Indicator	0.85	(4.03)***	0.82	(5.77)***	0.77	(5.42)***	0.80	(4.71)***
Corporate VC Indicator	0.78	(1.38)	0.67	(1.16)	0.66	(1.02)	0.71	(1.27)
Institutional VC Indicator	0.49	(5.19)***	0.40	(3.68)***	0.44	(4.09)***	0.48	(9.53)***
Early Investment Indicator	−0.23	(−3.59)***	−0.20	(−1.49)	−0.24	(−2.6)***	−0.23	(−1.74)*
Industry, year and round sequence fixed effects	Present		Present		Present		Present	
Log likelihood	972		968		969		970	
Number of observations	491		491		491		491	

As before, *Portfolio Company Quality*, *VC Industry Development*, *Founder Departure Indicator* and *Institutional VC indicator* all result in shorter *Investment Duration*, while the *Founder Overlap Indicator* predicts longer *Investment Duration*. In addition, domestic market conditions for IPO and M&A are insignificant for *Investment Duration*. One difference from the earlier results is that early VC investment predicts a longer *Investment Duration*. This may be explained by VC commitment: a VC firm willing to invest early may tend to be more committed to a portfolio company.

5. Analysis at the portfolio company level

In this section, we present performance analysis at the portfolio company level using logit and Cox hazard models. The unit of analysis is a portfolio company, with one observation for each portfolio company. The sample size is 243. We focus only on the initial VC funding for each portfolio company, since performance in later rounds may be heavily influenced by performance in earlier rounds. This sample selection may avoid the double counting problem in an analysis of VC investments. We identify the foreign lead VC firm and the CEO for each portfolio company at the initial VC funding, and take the foreign lead VC firm's human capital and domestic CEO experience as the primary explanatory variables. Here, the foreign lead VC firm is the VC firm that participated in the first round of financing and made the largest total investment in the portfolio company across all rounds.

Table 7 reports the results of our logit analysis on portfolio companies with *Investment Success* as the dependent variable. *Investment Success* takes the value of 1 if the portfolio company had gone public or had been acquired by September 2009 and 0 otherwise. We relate *Investment Success* to the same explanatory and control variables as before. All *t*-statistics listed in parentheses have been adjusted for heteroskedasticity (White, 1980) and clustered by industry and year (Petersen, 2009) accounting for correlations within an industry and in a year. The results in Table 7 are consistent with the earlier results on individual VC investments. Specifically, a one standard deviation increase in *CEO General Experience*, *Special Experience as CEO* or *Special Experience as Manager* increases the odds of *Investment Success* respectively by 22.3%, 17.3% and 19.2%. As earlier, foreign lead VCs' human capital has no significant effect on *Investment Success*. We also find a positive impact from *Portfolio Company Quality*, *VC Industry Development*, *Founder Departure Indicator*, and *Institutional VC Indicator*, and a negative impact from *Founder Overlap Indicator*. Again, we do not find a significant association between domestic market conditions and *Investment Success*.

Table 8 reports the results of our Cox hazard analysis on portfolio companies with *Investment Duration* as the dependent variable. *Investment Duration* of a successful portfolio company (that either went public or was acquired) is the logarithm of one plus the number of days from the date of its initial VC funding to the date of exit. It is right-censored at September 2009 for those portfolio companies that had not exited successfully by September 2009. As in the last section, CEO experience continues to be significantly effective in predicting *Investment Duration*. It has a large marginal effect. Specifically, a one standard deviation increase in *CEO General Experience*, *Special Experience as CEO* or *Special Experience as Manager* is associated with a 16.0%, 5.9% or 15.5% increase respectively in the hazard of a portfolio company's successful exit. Again, foreign lead VCs' human capital, such as experience, networks and reputation, is insignificant.

Taking the right-censored nature of the data into account, we further find that *Portfolio Company Quality*, *VC Industry Development*, *Founder Departure Indicator* and *Institutional VC Indicator* reduce *Investment Duration*, while the *Founder Overlap Indicator* raises *Investment Duration*.

In summary, our analysis of VC investments and portfolio companies in Tables 5–8 indicates that cross-border VC performance is largely determined by domestic entrepreneurs' experience. Foreign VCs' human capital plays little role on their performance in China. This is contrary to the existing literature that emphasizes VCs' human capital. The existing literature has seldom investigated VC performance from the contributions of entrepreneurs. Our findings are consistent with the recent findings in Gompers et al. (2010) and Chatterji (2009) in that entrepreneurs' experience helps their companies capture good opportunities and hence enhance performance.

6. Robustness checks

In the above analyses, CEO experience has emerged as a consistent predictor of the performance of portfolio companies and VC investments. In this section, we conduct many robustness tests to check

Table 7

Logit analysis of *Investment Success* at the portfolio company level. The sample consists of foreign VC-backed portfolio companies initially funded between 1999 and 2006 and that have relevant data in the databases. We trace the performance of the sample until September 2009, which allows at least 2.5 years for an exit. The sample is classified into two categories, one for successful exits (IPOs or acquisitions) and the other for unsuccessful exits (liquidations, including non-exits). The dependent variable is *Investment Success*, which takes the value of 1 if the portfolio company had gone public or had been acquired before September 2009 and 0 otherwise. The foreign lead VC firm's human capital includes three variables: *VC Experience*, *VC Networks*, and *VC Reputation*. Domestic entrepreneurs' experience includes three variables: *CEO General Experience*, *Special Experience as CEO*, and *Special Experience as Manager*. Other variables include *Syndicate Size*, *Portfolio Company Quality*, *VC Industry Development*, *Domestic IPO Conditions*, *Domestic M&A Conditions*, *Chairman Overlap Indicator*, *Founder Overlap Indicator*, *Founder Departure Indicator*, *Corporate VC Indicator*, *Institutional VC Indicator*, and *Early Investment Indicator*. The definitions, measures and data sources of the variables are described in Table 3. Industry and initial funding year fixed effects are included in all the regressions (not reported). Intercepts are not reported. Robust *t*-values, cluster-adjusted by industry and year, are in parentheses. The significance levels at 1%, 5% and 10% are identified by ***, ** and *, respectively.

	1		2		3		4	
	Coeff.	t-Value	Coeff.	t-Value	Coeff.	t-Value	Coeff.	t-Value
Foreign lead VCs' human capital								
VC Experience	−0.17	(−1.3)	−0.18	(−1.44)	−0.16	(−1.32)	−0.17	(−1.37)
VC Networks	−0.11	(−0.97)	−0.11	(−0.93)	−0.11	(−0.98)	−0.11	(−0.95)
VC Reputation	0.07	(0.43)	0.08	(0.46)	0.07	(0.45)	0.07	(0.41)
Domestic entrepreneurs' experience								
CEO General Experience			0.31	(3.09)***				
Special Experience as CEO					0.39	(4.41)***		
Special Experience as Manager							0.31	(1.83)*
Syndicate Size	0.03	(0.31)	0.02	(0.21)	0.01	(0.13)	0.01	(0.09)
Portfolio Company Quality	0.29	(2.63)***	0.29	(2.59)***	0.29	(2.58)***	0.29	(2.79)***
VC Industry Development	1.89	(4.94)***	1.86	(4.82)***	1.90	(5.29)***	1.92	(4.86)***
IPO and M&A market conditions								
Domestic IPO Conditions	−0.15	(−1.53)	−0.15	(−1.47)	−0.14	(−1.48)	−0.15	(−1.58)
Domestic M&A Conditions	0.63	(1.16)	0.60	(1.17)	0.64	(1.61)	0.67	(1.26)
Chairman Overlap Indicator	0.48	(1.52)	0.47	(1.49)	0.49	(1.7)*	0.49	(1.64)*
Founder Overlap Indicator	−2.68	(−2.53)**	−2.59	(−2.51)**	−2.69	(−2.59)***	−2.61	(−2.51)**
Founder Departure Indicator	2.08	(3.48)***	1.95	(3.39)***	2.06	(3.46)***	2.01	(3.33)***
Corporate VC Indicator	0.95	(1.1)	0.87	(1.01)	0.96	(1.14)	0.88	(1.06)
Institutional VC Indicator	0.77	(7.62)***	0.79	(4.75)***	0.82	(6.56)***	0.82	(8.4)***
Early Investment Indicator	0.07	(0.23)	0.11	(0.34)	0.12	(0.36)	0.10	(0.33)
Industry and year fixed effects	Present		Present		Present		Present	
Log likelihood	111		110		110		110	
Number of observations	241		241		241		241	
Pseudo-R ²	0.30		0.30		0.30		0.30	

reliability of our results. For brevity, we only report the results of the first two robustness tests on portfolio companies. The details of other tests are available on request.

6.1. IPOs versus acquisitions

In the above logit and Cox hazard analyses, we find that more experienced CEOs are associated with *Investment Success*. As a robustness check, we further investigate the likelihood of a portfolio company's exit via one of three major routes—IPOs, acquisitions or liquidations⁹—in a multinomial logit setting. Our analysis is to identify the determinants of exit choices of foreign VC-backed companies and to investigate differences in our measures of foreign lead VCs' human capital and domestic entrepreneurs experience on exit choices of IPOs versus acquisitions.

⁹ Liquidations include all exit routes other than IPOs and acquisitions, such as buyouts, secondary sales and write-offs. As mentioned at the beginning of Section 3, since we trace at least 2.5 years for an exit, those investments that had not exited by September 2009 (non-exits) make up only a very small proportion of our sample. Due to data limitations, we cannot identify these non-exits clearly, and hence also classify them as liquidations. A similar approach is taken by Gompers and Lerner (2000), Hochberg et al. (2007), Gompers et al. (2008), and Nahata (2008).

Table 8

Cox hazard analysis of *Investment Duration* at the Portfolio Company Level. The sample consists of foreign VC-backed portfolio companies initially funded between 1999 and 2006 and that have relevant data in the databases. We trace the performance of the sample until September 2009, which allows at least 2.5 years for an exit. The sample is classified into two categories, one for successful exits (IPOs or acquisitions) and the other for unsuccessful exits (liquidations, including non-exits). The Cox hazard model is used and the dependent variable is *Investment Duration*. *Investment Duration* of successful portfolio companies (that either went public or were acquired) is the logarithm of one plus the number of days from the date of its initial VC funding to the date of exit. It is right-censored at September 2009 for portfolio companies that had not exited successfully by September 2009. A positive (negative) coefficient in the Cox hazard model indicates that the covariate increases (decreases) the hazard and shortens (lengthens) the expected duration. The foreign lead VC firm's human capital includes three variables: *VC Experience*, *VC Networks*, and *VC Reputation*. Domestic entrepreneurs' experience includes three variables: *CEO General Experience*, *Special Experience as CEO*, and *Special Experience as Manager*. Other variables include *Syndicate Size*, *Portfolio Company Quality*, *VC Industry Development*, *Domestic IPO Conditions*, *Domestic M&A Conditions*, *Chairman Overlap Indicator*, *Founder Overlap Indicator*, *Founder Departure Indicator*, *Corporate VC Indicator*, *Institutional VC Indicator*, and *Early Investment Indicator*. The definitions, measures and data sources of the variables are described in Table 3. Industry and initial funding year fixed effects are included in all the regressions (not reported). Intercepts are not reported. Robust *t*-values, cluster-adjusted by industry and year, are in parentheses. The significance levels at 1%, 5% and 10% are identified by ***, ** and *, respectively.

	1		2		3		4	
	Coeff.	t-Value	Coeff.	t-Value	Coeff.	t-Value	Coeff.	t-Value
Foreign lead VCs' human capital								
VC Experience	−0.18	(−0.95)	−0.20	(−1.18)	−0.18	(−1.01)	−0.19	(−1.03)
VC Networks	−0.05	(−0.54)	−0.05	(−0.5)	−0.05	(−0.56)	−0.05	(−0.53)
VC Reputation	0.04	(0.25)	0.05	(0.31)	0.05	(0.3)	0.04	(0.27)
Domestic Entrepreneurs' Experience								
CEO General Experience			0.23	(4.82)***				
Special Experience as CEO					0.14	(1.7)*		
Special Experience as Manager							0.26	(3.26)***
Syndicate Size	0.01	(0.19)	0.01	(0.15)	0.01	(0.11)	0.00	(−0.03)
Portfolio Company Quality	0.20	(3.16)***	0.20	(2.8)***	0.20	(2.91)***	0.20	(3.01)***
VC Industry Development	0.55	(2.3)**	0.49	(2.08)**	0.55	(2.3)**	0.54	(2.23)**
IPO and M&A market conditions								
Domestic IPO Conditions	−0.15	(−1.49)	−0.14	(−1.27)	−0.14	(−1.35)	−0.14	(−1.32)
Domestic M&A Conditions	0.25	(0.95)	0.20	(0.8)	0.24	(0.89)	0.24	(0.92)
Chairman Overlap Indicator	0.33	(1.42)	0.32	(1.43)	0.34	(1.52)	0.36	(1.59)
Founder Overlap Indicator	−1.09	(−2.99)***	−1.08	(−3.13)***	−1.07	(−3.09)***	−1.01	(−3.21)***
Founder Departure Indicator	0.66	(3.79)***	0.63	(5.35)***	0.63	(3.93)***	0.59	(4.41)***
Corporate VC Indicator	0.70	(1.39)	0.63	(1.3)	0.68	(1.35)	0.67	(1.35)
Institutional VC Indicator	0.72	(1.71)*	0.67	(1.66)*	0.69	(1.62)	0.72	(1.97)**
Early Investment Indicator	−0.20	(−0.99)	−0.17	(−0.97)	−0.20	(−0.98)	−0.22	(−1.1)
Industry and year fixed effects	Present		Present		Present		Present	
Log likelihood	380		379		380		379	
Number of observations	239		239		239		239	

IPOs and acquisitions are the two primary methods by which privately-held firms become public companies. Bayar and Chemmanur (2009) investigated theoretically the exit strategies chosen by entrepreneurs and VCs, focusing on IPOs and acquisitions as the two most profitable exit routes. Studies by Brau et al. (2003) and Poulsen and Stegemoller (2008) have identified factors that determine a firm's choice between IPO and acquisition using a broad sample of privately-held firms. However, few scholars have empirically investigated the exit choices of foreign VC-backed companies.

Table 9 reports the results of our multinomial logit regressions on our measures of foreign lead VCs' human capital, domestic entrepreneurs' experience, and other control variables used in our earlier analyses. As shown, the main results remain unchanged. We further find that the effects of CEO experience on the choices between IPOs and acquisitions are comparable. For example, model 1 indicates that a one standard deviation increase in *CEO General Experience* raises the odds of an IPO exit by 22.2% and the odds of an acquisition by 28.0%. The findings also point to the importance of domestic entrepreneurs' experience to the exits by IPOs and acquisitions. Further, Table 9 shows that better *VC Industry Development*, a departure of the founder and a separation of CEO and founder all improve the likelihood of an IPO, and also the likelihood of an acquisition to a similar extent. Table 9 also shows some different effects of the control

Table 9

Multinomial logit analysis of IPO and M&A exits at the portfolio company level. The sample consists of foreign VC-backed portfolio companies initially funded between 1999 and 2006 and that have relevant data in the databases. We trace the performance of the sample until September 2009, which allows at least 2.5 years for an exit. The sample is classified into two categories, one for successful exits (IPOs or acquisitions) and the other for unsuccessful exits (liquidations, including non-exits). All the models are multinomial logit models. The dependent variable takes three discrete values respectively for IPOs, acquisitions, and liquidations (including non-exits). The foreign lead VC firm's human capital includes three variables: *VC Experience*, *VC Networks*, and *VC Reputation*. Domestic entrepreneurs' experience includes three variables: *CEO General Experience*, *Special Experience as CEO*, and *Special Experience as Manager*. Other variables include *Syndicate Size*, *Portfolio Company Quality*, *VC Industry Development*, *Domestic IPO Conditions*, *Domestic M&A Conditions*, *Chairman Overlap Indicator*, *Founder Departure Indicator*, *Corporate VC Indicator*, *Institutional VC Indicator*, and *Early Investment Indicator*. The definitions, measures and data sources of all variables are described in Table 3. Industry and initial funding year fixed effects are included in all the regressions (not reported). Intercepts are not reported. Robust *t*-values, cluster-adjusted by industry and year, are in parentheses. The significance levels at 1%, 5% and 10% are identified by ***, **, and *, respectively.

	1			2			3		
	IPO	M&A		IPO	M&A		IPO	M&A	
	Coeff.	<i>t</i> -Value	Coeff.	Coeff.	<i>t</i> -Value	Coeff.	Coeff.	<i>t</i> -Value	<i>t</i> -Value
Foreign lead VCs' human capital									
VC experience	−0.41	(−0.86)	0.23	−0.38	(−0.42)	0.25	−0.39	(−1.2)	0.24
VC Networks	−0.04	(−0.51)	−0.13	−0.05	(−0.51)	−0.12	−0.04	(−0.52)	−0.12
VC Reputation	−0.24	(−0.06)	0.26	−0.25	(−0.04)	0.25	−0.25	(−0.08)	0.24
Domestic Entrepreneur's Experience									
CEO General Experience	0.30	(2.07)**	0.36	0.35	(2.28)**	0.48			
Special Experience as CEO							0.29	(1.69)*	0.37
Special Experience as Manager							0.15	(4.78)***	−0.16
Syndicate Size	0.16	(6.95)***	−0.16	0.16	(3.42)***	−0.16	0.15	(3.04)***	−0.08
Portfolio Company Quality	0.39	(2.91)***	−0.10	0.39	(2.14)**	−0.08	0.39	(3.04)***	−0.08
VC Industry Development	2.03	(12.57)***	2.28	2.07	(16.28)***	2.28	2.07	(9.31)***	2.36
IPO and M&A market conditions									
Domestic IPO Conditions	−0.16	(−0.72)	−0.16	−0.15	(−0.5)	−0.12	−0.17	(−0.69)	−0.12
Domestic M&A Conditions	0.42	(0.64)	0.99	0.46	(0.71)	1.02	0.47	(0.72)	1.09
Chairman Overlap Indicator	0.65	(1.49)	0.33	0.66	(2.4)**	0.29	0.67	(1.54)	0.30
Founder Overlap Indicator	−3.16	(−2.5)**	−2.37	−3.26	(−2.44)**	−2.52	−3.19	(−2.32)**	−2.46
Founder Departure Indicator	2.30	(3.46)***	2.02	2.41	(3.39)***	2.13	2.36	(3.32)***	2.10
Corporate VC Indicator	1.75	(2.68)***	−0.39	1.86	(3.02)***	−0.28	1.78	(2.86)***	−0.38
Institutional VC Indicator	1.67	(3.43)***	−1.22	1.70	(3.3)***	−1.24	1.70	(3.5)***	−1.22
Early Investment Indicator	−0.38	(−1.56)	0.55	−0.37	(−1.17)	0.56	−0.39	(−1.44)	0.54
Industry and year fixed effects	Present			Present			Present		
Log likelihood	136			136			136		
Number of observations	242			242			242		
Pseudo-R ²	0.37			0.37			0.37		

variables, such as total VC funding, syndicate size, a corporate lead VC firm, an institutional lead VC firm and early VC investment, on exits by IPOs versus acquisitions.

6.2. An instrumental variable approach to correct for endogeneity

Endogeneity is often a major concern in studies of corporate finance. In our case, the question is whether the observed positive correlation between domestic CEO experience and VC performance may also be due to a “selection” effect by which successful companies are more likely to match with experienced CEOs. If so, the match between a portfolio company and its CEO may to some extent be endogenous, and is likely to be related to company characteristics. If this match is non-random, it is possible that the independent variables are correlated with the regression error term, leading to a biased estimation. In the earlier analysis, we try to capture the quality of the portfolio companies using the total VC funding across all rounds. As an additional robustness check and to address the potential endogeneity problem, we use the classic instrumental variable approach. Recent studies that use this approach in corporate finance issues include those of [Berger et al. \(2005\)](#), [Faulkender and Petersen \(2005\)](#), [Hellmann et al. \(2008\)](#) and [Bottazzi et al. \(2008\)](#).

In our case, a proper instrumental variable must be independent of the outcome equation but related to domestic CEO experience. The common choices for instruments are measures of locally available characteristics, such as CEO experience in our case. The main idea is that the *Local Availability of Experienced CEOs* affects the likelihood of a company's matching with a certain CEO; but once a match is made, it should not affect how the specific company and the specific CEO interact. In other words, although a company's actual choice of CEO may be endogenous, the *Local Availability of Experienced CEOs* is exogenous. Moreover, once a company is matched with a CEO, the *Local Availability of Experienced CEOs* becomes irrelevant, since all that matters is the experience of the chosen CEO. To deal with this issue, we define a variable called *Local Availability of Experienced CEOs*, which is a fraction where the denominator is the number of portfolio companies in our sample that are located in the same province as the subject portfolio company and the numerator is the number of portfolio companies that have hired CEOs who have previously worked for at least one other company whatever the job. This definition mirrors the approach of [Berger et al. \(2005\)](#) and [Bottazzi et al. \(2008\)](#).¹⁰

We employ a two-step procedure to correct for endogeneity. In the first step, we estimate *CEO General Experience* using the instrumental variable *Local Availability of Experienced CEOs* and other covariates in an OLS model. In the second step, we use the estimates of *CEO General Experience* from the first step to estimate each of the following measures of company performance: *Investment Success*, *Investment Duration*, and *IPOs versus acquisitions*, using logit, Cox hazard and multinomial logit models, respectively.

[Table 10](#) presents the regression results. As shown in model 1, *Local Availability of Experienced CEOs* is highly significant and has the expected positive sign in predicting *CEO General Experience*, which makes it a relevant instrument. The second-step regressions in models 2–4 indicate that accounting for potential endogeneity does not change the statistical significance of *CEO General Experience*. In fact, adding the selection equation hardly affects the main model at all.

As a further robustness test, we also define *Local Availability of Experienced CEOs* based respectively on CEOs' *Special Experience as CEO* and as top manager. The results remain the same.

6.3. Alternative measures of foreign VCs' human capital

An inappropriate measure of foreign VCs' human capital could be the cause of insignificance of foreign VCs' human capital on VC performance. To rule out this possibility, we test several other measures of foreign VCs' human capital, where human capital includes experience, networks and reputation.

[Sorensen \(2007\)](#) argues that the cumulative number of investment rounds made by a VC firm is a better measure of experience than the VC firm's age, since the age cannot properly distinguish between active and inactive VCs. Hence, we construct an alternative measure of *VC Experience* as the cumulative number of

¹⁰ [Berger et al. \(2005\)](#) choose the median size of banks in the local market as an instrumental variable in estimating individual bank sizes. [Bottazzi et al. \(2008\)](#) choose the local availability of experienced investors as an instrumental variable in estimating investors' experience.

Table 10

Portfolio companies' performance after controlling for endogeneity. The sample consists of foreign VC-backed portfolio companies initially funded between 1999 and 2006 and that have relevant data in the databases. We trace the performance of the sample until September 2009, which allows at least 2.5 years for an exit. The sample is classified into two categories, one for successful exits (IPOs or acquisitions) and the other for unsuccessful exits (liquidations, including non-exits). The first model is a selection equation (OLS model) with *CEO General Experience* as the dependent variable. The instrumental variable for *CEO General Experience* in the first-step selection equation is *Local Availability of Experienced CEOs*. The dependent variable in the second model (logit model) is *Investment Success*, which takes the value of 1 if the portfolio company had gone public or had been acquired before September 2009 and 0 otherwise. The dependent variable of the third model (Cox hazard model) is *Investment Duration*. *Investment Duration* of a successful portfolio company (that either went public or was acquired) is the logarithm of one plus the number of days from the date of its initial VC funding to the date of exit. It is right-censored at September 2009 for portfolio companies that had not exited successfully by September 2009. A positive (negative) coefficient in the Cox hazard model indicates that the covariate increases (decreases) the hazard and shortens (lengthens) the expected duration. The dependent variable of the fourth model (multinomial logit model) takes three discrete values respectively for IPOs, acquisitions, and liquidations (including non-exits). The foreign lead VC firm's human capital includes three variables: *VC Experience*, *VC Networks*, and *VC Reputation*. Domestic entrepreneurs' experience in the second step is the predicted value generated from the first step of the OLS estimation. Other variables include *Syndicate Size*, *Portfolio Company Quality*, *VC Industry Development*, *Domestic IPO Conditions*, *Domestic M&A Conditions*, *Chairman Overlap Indicator*, *Founder Overlap Indicator*, *Founder Departure Indicator*, *Corporate VC Indicator*, *Institutional VC Indicator*, and *Early Investment Indicator*. The definitions, measures and data sources of the variables are described in Table 3. Industry and initial funding year fixed effects are included in all the regressions (not reported). Intercepts are not reported. Robust *t*-values, cluster-adjusted by industry and year, are in parentheses. The significance levels at 1%, 5% and 10% are identified by ***, ** and *, respectively.

	First step		Second step							
	1		2		3		4			
	OLS		Logit		Cox hazard		Multinomial logit			
							IPO		M&A	
	Coeff.	t-Value	Coeff.	t-Value	Coeff.	t-Value	Coeff.	t-Value	Coeff.	t-Value
Foreign lead VCs' human capital										
VC Experience	0.02	(0.5)	−0.15	(−0.77)	−0.20	(−1.47)	−0.36	(−1.39)	0.23	(1.12)
VC Networks	0.001	(0.03)	−0.11	(−2.36)	−0.05	(−1.64)	−0.05	(−0.34)	−0.12	(−1.57)
				**						
VC Reputation	−0.01	(−0.32)	0.06	(0.78)	0.05	(0.48)	−0.25	(−0.8)	0.26	(1.01)
Domestic entrepreneur's experience										
CEO General Experience			0.27	(1.81)*	0.48	(2.45)**	0.45	(1.48)	0.64	(1.89)*
Syndicate Size	0.02	(0.71)	0.04	(0.46)	0.006	(0.07)	0.19	(2.01)**	−0.15	(−1.14)
Portfolio Company Quality	0.01	(0.36)	0.30	(1.9)*	0.20	(1.96)**	0.40	(2.53)**	−0.11	(−1.35)
VC Industry Development	0.15	(1.21)	1.93	(5.16)	0.49	(1.45)	2.13	(5.58)	2.19	(7.45)
				***				***		***
IPO and M&A market conditions										
Domestic IPO Conditions	0.04	(0.78)	−0.14	(−0.94)	−0.17	(−1.92)	−0.16	(−1.05)	−0.14	(−0.29)
						*				
Domestic M&A Conditions	0.11	(1.25)	0.66	(2.22)**	0.21	(0.73)	0.51	(2.82)	0.96	(3.21)
								***		***
Chairman Overlap Indicator	0.09	(1.03)	0.50	(1.63)	0.28	(0.88)	0.68	(2.3)**	0.23	(0.48)
Founder Overlap Indicator	−0.20	(−1.2)	−2.76	(−8.03)	−0.99	(−2.63)	−3.40	(−2.14)	−2.46	(−1.95)
				***		***		**		*
Founder Departure Indicator	0.29	(1.86)**	2.17	(6.06)	0.52	(1.17)	2.59	(2.12)**	2.00	(1.74)*

Corporate VC Indicator	0.26	(1.66)*	1.02	(2.15)**	0.59	(1.71)*	1.98	(1.55)	−0.43	(−0.56)
Institutional VC Indicator	0.07	(0.41)	0.78	(1.2)	0.75	(2.05)**	1.66	(3.64)	−1.22	(−5.61)
								***		***
Early Investment Indicator	−0.23	(−2.49)	0.03	(0.06)	−0.10	(−0.32)	−0.47	(−1.33)	0.59	(1.33)

Local Availability of Experienced CEOs	1.25	(4.33)								

Industry and year fixed effects	Present		Present		Present		Present			
Log likelihood			111		380		136			
Number of observations	242		241		239		242			
R ² or Pseudo-R ²	0.22		0.30				0.37			

investment rounds. Following [Hochberg et al. \(2007\)](#), we construct two alternative measures of foreign VCs' networks. Given a VC firm, one measure is the normalized number of VCs whom a VC firm has invited to its syndicate, and the other is the normalized number of VCs who have invited the VC firm to their syndicates. Following [Nahata \(2008\)](#), we construct an alternative short-term measure of foreign VCs' reputation as a VC firm's dollar capitalization share of the IPOs backed by the VC firm, relative to other VCs in the previous 5 years on a rolling basis in VC markets worldwide. As before, foreign VCs' human capital is measured at the time of a VC firm's first investment into a portfolio company. We re-run all our regressions using these alternative measures. We find that all our results remain qualitatively the same.

We also use the average foreign VCs' human capital to replace the lead VC firm's human capital at the time of a portfolio company's receiving initial VC funding in our analysis of portfolio companies. Again, our results remain the same.

6.4. Further robustness checks

We have also done several other robustness tests. We here report a summary of these tests.

First, considering the current situation in China's VC industry and private companies, we introduce a different definition of success. In China, private companies are desperately short of funding. Their internal financing resources are limited. External financing sources such as banks are reluctant to lend to them because of high risks. China's VC industry is still far from adequately developed to satisfy private companies' needs. Hence, trying to obtain funding is essential to most private companies. To some extent, receiving later funding successfully implies success in earlier operations. Hence, we introduce a second definition of success, which equals 1 if the portfolio company went public, or was acquired, or received follow-up funding before September 2009, and 0 otherwise. The likelihood of success increases to 66% after this adjustment. The main results are very similar to that reported earlier, suggesting that our findings are robust to changes in the measure of success.

Second, following [Hochberg et al. \(2007\)](#), we construct a different indicator at the company level as the dependent variable, which equals 1 if the company had survived a financing round to receive another financing round or exit via an IPO or M&A transaction, and 0 otherwise. The sample size decreases from round to round. We use the same explanatory variables as before. The main results remain the same.

Third, following [Kannianen and Keuschnigg's \(2003, 2004\)](#) theoretical studies and [Cumming's \(2006\)](#) empirical work, we further control for a VC firm's portfolio size measured by the number of companies in the VC firm's worldwide portfolio. We find a significant negative relationship between portfolio size and VC performance, while our main results remain qualitatively unaltered. This finding is quite consistent with the findings of [Kannianen and Keuschnigg \(2003, 2004\)](#) and [Cumming \(2006\)](#) that there is a tradeoff between portfolio size and the intensity of managerial advice.

Fourth, to investigate further whether or not our results are unduly influenced by outliers, we re-run our regressions after winsorizing the top and bottom 1%, 2% and 5%, respectively, for each variable. The results remain largely unchanged.

Fifth, considering the possible impact of the NASDAQ bubble during 1999–2000, we re-run our regressions using a subsample which excludes those 2 years. The sample size is reduced to 420 and 205 for the analyses of VC investments and portfolio companies, respectively. Our results are qualitatively unchanged.

Finally, as shown in Panel B of [Table 2](#), the computer related and biotechnology industries make respectively the largest and smallest contributions in our dataset. To examine whether or not the inclusion of these industries in our sample influences our results, we re-run our regressions without these industries. The results remain qualitatively unchanged.

7. Conclusions

This study investigates the determinants of cross-border VC performance in the Chinese market. Different from the existing studies, we investigate the characteristics of both VCs and entrepreneurs on VC performance. Our findings provide much interesting evidence on cross-border VC performance, which enriches the literature on cross-border VC investments.

Using 495 VC investments by 84 foreign VCs in 243 Chinese domestic companies, we analyze the roles of both foreign VCs' human capital and domestic entrepreneurs' experience on *Investment Success* and *Investment Duration* after controlling for *Portfolio Company Quality*, domestic VC *Industry Development*, exit market conditions, and a number of other factors. We find that foreign VCs' human capital from VC markets worldwide has little impact on their cross-border VC performance in China. This is inconsistent with the existing literature on VC performance, which generally finds that VCs' human capital is essential to their domestic investment performance. Our finding suggests that VCs' human capital is market-specific. Further, we find that domestic entrepreneurs' experience plays an important role in *Investment Success* and *Investment Duration*. Specifically, if a CEO has more general experience, special experience as CEO or special experience as top manager, *Investment Success* is significantly increased and *Investment Duration* is significantly shortened. In addition, we find that *Investment Success* and *Investment Duration* are significantly related to a departure of the founder, a separation of CEO and founder, the portfolio company's quality and China's VC industry development, while it is little related to China's market conditions for IPO and M&A.

This paper has two main contributions. Firstly, our analysis helps elucidate the factors underlying cross-border VC performance in emerging markets. In recent years, cross-border VC investment has become a significant phenomenon. Such investment can have an important impact on emerging markets in terms of innovation, technology transfer and supporting private companies. There is an expanding literature on VC in Asia and on cross-border VC investments, but there is still a lack of understanding of the factors influencing the performance of cross-border VC investments. The Chinese VC market has grown rapidly during this decade and China has now become one of the main VC importers. This study was designed to lead the way in the investigation of cross-border VC performance in China.

Secondly, this study deals with the roles of both VCs and entrepreneurs on VC performance. The literature on VC performance generally focuses on the role of VC firms but pays little attention to the role of entrepreneurs. Departing from the findings of previous studies, we find that VCs' human capital has little relevance to its success in China. Instead, we find that local entrepreneurs' experience is much more important.

Due to limited availability of data, there are some weaknesses in our analysis. The sample size is relatively small and the time span is relatively short. The variables used in the analyses may have been too limited. For example, we do not have complete data on entrepreneurs' level of education and VCs' China experience. We also lacked data to examine the determinants of long-term performance. As richer data becomes available in the future after VC firms have built a larger presence in China, we will be able to enrich our research.

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